CHMB23 · Thermodynamics Ch. 1-3 · Formula Sheet

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Ideal gas (n **mol, const** $C_{V,m}$; $C_{p,m} = C_{V,m} + R$) | **state functions** $\Delta U, \Delta H, \Delta S$ | **path functions** q, w

Process	w (on gas)	q	ΔU	ΔH	ΔS_{sys}	ΔS_{surr} (reservoir)	ΔS_{tot}	
Isothermal, rev	$-nRT\ln(V_f/V_i) = -nRT\ln(p_i/p_f)$	$-w$	0	0	$nR\ln(V_f/V_i)$	$-nR\ln(V_f/V_i)$	0	$W_{out} = W_{out,max}$
Isothermal, const p_{ex}	$-p_{ex}(V_f - V_i)$	$-w$	0	0	$nR\ln(V_f/V_i)$	$-q/T$	> 0	$W_{out} < W_{out,rev}$
Free expansion ($p_{ex} = 0$, **insulated**)	0	0	0	0	$nR\ln(V_f/V_i)$	0	> 0	
Isochoric	0	$nC_{V,m}\Delta T$	$nC_{V,m}\Delta T$	$nC_{p,m}\Delta T$	$nC_{V,m}\ln(T_f/T_i)$	$-q/T_{surr}$	≥ 0	$q_V = \Delta U$; $\Delta H - \Delta U = nR\Delta T$
Isobaric	$-p\Delta V = -nR\Delta T$	$nC_{p,m}\Delta T$	$nC_{V,m}\Delta T$	$nC_{p,m}\Delta T$	$nC_{p,m}\ln(T_f/T_i)$	$-q/T_{surr}$	≥ 0	$q_p = \Delta H$
Adiabatic, rev	$nC_{V,m}\Delta T$	0	$nC_{V,m}\Delta T$	$nC_{p,m}\Delta T$	0	0	0	$TV^{\gamma-1} = pV^\gamma = \text{const}$, $\gamma = C_p/C_V$
Phase change at T_{trs}, **const** p	$-p\Delta V$	$\Delta_{trs}H$	$\Delta_{trs}H - p\Delta V$	$\Delta_{trs}H$	$\Delta_{trs}H/T_{trs}$	$-\Delta_{trs}H/T_{trs}$	0	$\Delta H_{freeze} = -\Delta_{fus}H$

Constants · units · signs 常数与约定

- $R = 8.314\ \text{J K}^{-1}\text{mol}^{-1} = N_A k_B$ | $k_B = 1.380649 \times 10^{-23}\ \text{J K}^{-1}$ | $N_A = 6.022 \times 10^{23}\ \text{mol}^{-1}$
- $pV = nRT = Nk_BT$ | $N = nN_A$ | $RT_{298.15} = 2.479\ \text{kJ mol}^{-1}$
- $R\ln 2 = 5.76$ | $R\ln\frac{3}{2} = 3.37$ | $R\ln 3 = 9.13\ \text{J K}^{-1}\text{mol}^{-1}$
- $T/\text{K} = \theta/°\text{C} + 273.15$ | $\Delta T/\text{K} = \Delta\theta/°\text{C}$ | $T_{tp}(\text{H}_2\text{O}) = 273.16\ \text{K}$
- $p^\circ = 1\ \text{bar} = 10^5\ \text{Pa}$ | $1\ \text{atm} = 101.325\ \text{kPa} = 1.01325\ \text{bar}$
- **SATP** 298.15 K, 1 bar: $V_m = RT/p = 24.79\ \text{L mol}^{-1}$
- **STP** 273.15 K, 1 atm: $V_m = 22.41\ \text{L mol}^{-1}$
- **Signs** $\Delta U = q + w$ | $q > 0$ in | $w > 0$ on system | $W_{out} = -w$
- **Units** $M\ [\text{kg mol}^{-1}]$ | $C\ [\text{J K}^{-1}]$ | $C_m,\ S_m\ [\text{J K}^{-1}\text{mol}^{-1}]$ | T in K

Work 功

- $\delta w = -p_{ex}\,dV$ (rev & irrev) | $w = -\int_{V_i}^{V_f} p_{ex}\,dV$
- **Const** p_{ex} $w = -p_{ex}(V_f - V_i)$ | **vacuum** $p_{ex} = 0 \Rightarrow w = 0$
- **Rev isothermal** $w = -nRT\ln\frac{V_f}{V_i} = -nRT\ln\frac{p_i}{p_f}$ | $p_1V_1 = p_2V_2$
- $W_{out} = \int p_{ex}\,dV$ (area under p-V) | $W_{out} \leq W_{out,rev} = \int p_{gas}\,dV$
- 1 mol, 300 K, $V \to 2V$: $W_{out,rev} = RT\ln 2 = 1.73\ \text{kJ}$
- $\ln(1+x) \approx x\ (x \ll 1)$

First law · enthalpy · calorimetry 第一定律与焓

- $\Delta U = q + w$ | **isolated** $q = w = 0 \Rightarrow \Delta U = 0$ | $E = U + E_{kin} + E_{pot}$
- **State** U, H, S, A, G: $\oint dX = 0$ | **path** $\delta q, \delta w$ | $q_1 + w_1 = q_2 + w_2$
- **Cycle** $\Delta U = 0$ | $q_{net} = -w_{net}$
- **Ideal gas** $U = U(T)$ | $\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p = 0$ | $U_{mono} = \frac{3}{2}nRT$
- $H \equiv U + pV = G + TS$ | $dH = dU + p\,dV + V\,dp$ | $\Delta H = \Delta U + \Delta(pV)$
- **Const** V $q_V = \Delta U \neq \Delta H$ | $\Delta H = q_V + V\Delta p$
- **Const** p $q_p = \Delta H$ (pV work only) | $\Delta H = q_p + w_{other}$
- **Bomb (const** V**)** $q_{cal} = C_{cal}\Delta T$ | $\Delta U_{rxn} \approx -C_{cal}\Delta T$ | **open (const** p**)** ΔH
- **Exo** $\Delta H < 0$ | **endo** $\Delta H > 0$ | $q_{surr} = -q_{sys}$ | $\Delta H_{rev.\,rxn} = -\Delta H$
- $\Delta_r H = \Delta_r U + \Delta\nu_g RT$ | $\Delta\nu_g = \nu_{gas,prod} - \nu_{gas,react}$ | $A(g) \to 2B(g)$: $\Delta_r H - \Delta_r U = RT$

Heat capacity · equipartition 热容

- $C_V \equiv \left(\frac{\partial U}{\partial T}\right)_V$ | $C_p \equiv \left(\frac{\partial H}{\partial T}\right)_p$ | $\Delta U = \int C_V dT$ | $\Delta H = \int C_p dT$
- $C = nC_m$ (extensive vs intensive) | $\gamma = C_p/C_V$
- **Ideal** $C_p - C_V = nR$ | $C_{p,m} - C_{V,m} = R$ | **general** $C_p - C_V = \frac{TV\alpha^2}{\kappa_T} \geq 0$
- **Equipartition** $\frac{1}{2}k_BT$ per quadratic term | $U_{m,trans} = \frac{3}{2}RT$ | $U_{m,rot} = RT$ (linear), $\frac{3}{2}RT$ (nonlinear)
- $C_{V,m}$: $\frac{3}{2}R$ (monatomic; $C_{p,m} = \frac{5}{2}R, \gamma = \frac{5}{3}$) | $\frac{5}{2}R$ (linear) | $3R$ (nonlinear, no vib)
- **Validity** $k_BT \gg \Delta\varepsilon$ | $\langle\varepsilon_{vib}\rangle = \frac{1}{2}\hbar\omega + \frac{\hbar\omega}{e^x - 1}, x = \frac{\hbar\omega}{k_BT}$
- **Kirchhoff** $\Delta_r H^\circ(T') = \Delta_r H^\circ(T) + \int_T^{T'} \Delta_r C_p^\circ\,dT \approx \Delta_r H^\circ(T) + \Delta_r C_p^\circ(T' - T)$
- $\Delta_r C_p^\circ = \sum \nu C_{p,m}^\circ(\text{prod}) - \sum \nu C_{p,m}^\circ(\text{react})$
- **Adiabatic** $q = 0$ | $\Delta U = w = nC_{V,m}\Delta T$ | **rev** $TV^{\gamma-1} = \text{const}, pV^\gamma = \text{const}$
- **Debye** ($T \ll \Theta_D$) $C_{V,m} \approx aT^3 = \frac{12\pi^4 R}{5}\left(\frac{T}{\Theta_D}\right)^3$ | $T \to 0$: $C \to 0$ | metals $+\gamma_e T$

Kinetic theory · gases · Boltzmann 动理论

- $pV = \frac{1}{3}Nm\langle v^2\rangle$ | $\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}k_BT$ | $\frac{1}{2}M\langle v^2\rangle = \frac{3}{2}RT$
- $v_{rms} = \sqrt{\frac{3RT}{M}}$ | $\bar{v} = \sqrt{\frac{8RT}{\pi M}}$ | $v_{mp} = \sqrt{\frac{2RT}{M}}$ | $v \propto \sqrt{T/M}$
- $v_{mp} : \bar{v} : v_{rms} = 1 : 1.128 : 1.225$ | N_2, 300 K: $v_{rms} \approx 517\ \text{m s}^{-1}$
- **Maxwell** $f(v) \propto v^2 e^{-Mv^2/2RT}$ | $\int_0^\infty f\,dv = 1$ | $T\uparrow$: peak $\to$, broader, lower
- **Boltzmann** $\frac{N_j}{N_i} = \frac{g_j}{g_i}e^{-(\varepsilon_j - \varepsilon_i)/k_BT}$ | $\frac{d}{dT}\ln\frac{N_j}{N_i} = \frac{\varepsilon_j - \varepsilon_i}{k_BT^2} > 0$
- $p_J \equiv x_J p$ | $x_J = n_J/\sum n_K$ | $\sum p_J = p$ | **ideal** $p_J = n_J RT/V$
- $Z = \frac{pV_m}{RT} = 1 + \frac{B}{V_m} + \frac{C}{V_m^2} + \cdots$ | $p \to 0$ (const T): $Z \to 1$
- **Mech. eq.** $p_1 = p_2$ | (same T) $V_1/V_2 = n_1/n_2$ | **thermal** $T_1 = T_2$ | **chem.** $\mu_1 = \mu_2$
- **Ideal mixture** $\Delta_{mix}H = \Delta_{mix}V = 0$ | $\Delta_{mix}S = -nR\sum x_i\ln x_i$ | $\mu_i = \mu_i^* + RT\ln x_i$
- **Box** $E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2) \propto V^{-2/3}$ | $q_{trans} = \frac{V}{\Lambda^3}$, $\Lambda = \frac{h}{\sqrt{2\pi m k_B T}}$

Entropy · second law 熵与第二定律

- **Definition** $dS = \delta q_{rev}/T$ | $\Delta S = \int_i^f \delta q_{rev}/T$ (rev path, same endpoints)
- **Clausius** $dS \geq \delta q/T$ (= rev, > irrev) | **isolated** $\Delta S = S_{gen} \geq 0$
- **Balance** $\Delta S = \int \delta q/T_b + S_{gen}$ | $S_{gen} \geq 0$ | rev adiabatic: $\Delta S = 0$
- **Boltzmann** $S = k_B\ln W$ | $W = 1 \Rightarrow S = 0$ | **Gibbs** $S = -k_B\sum p_i\ln p_i$
- $\Delta S_{sys} = \int \delta q_{rev}/T$ | $\Delta S_{surr} = -q_{sys}/T_{surr}$ | $\Delta S_{tot} = \Delta S_{sys} + \Delta S_{surr} \geq 0$
- **Const** T, p $\Delta S_{surr} = -\Delta H_{sys}/T$
- **Hot → cold** $\Delta S_{tot} = Q\left(\frac{1}{T_c} - \frac{1}{T_h}\right) > 0$
- $\Delta S = q/T$: 100 J → 1 J K^{-1} (100 K), 0.1 J K^{-1} (1000 K)
- **Spontaneous** $\Delta S_{tot} > 0$ ($\Delta S_{sys} < 0$ allowed)

ΔS calculations 熵变计算

- **Ideal gas** $\Delta S = nR\ln\frac{V_f}{V_i} + nC_{V,m}\ln\frac{T_f}{T_i} = -nR\ln\frac{p_f}{p_i} + nC_{p,m}\ln\frac{T_f}{T_i}$
- **Isothermal** $\Delta S_m = R\ln\frac{V_f}{V_i}$ | $V \times 2$: $+5.76$ | $V \times \frac{1}{2}$: $-5.76\ \text{J K}^{-1}\text{mol}^{-1}$
- **Heating** $\Delta S = \int_{T_i}^{T_f}\frac{C_p}{T}dT = C_p\ln\frac{T_f}{T_i}$ (const p) | C_V (const V) | $q = C\Delta T$
- **Transition** $\Delta_{trs}S = \frac{\Delta_{trs}H}{T_{trs}}$ | $\Delta_{fus}S = \frac{\Delta_{fus}H}{T_m}$ | $\Delta_{vap}S = \frac{\Delta_{vap}H}{T_b}$
- H_2O freezing, 1 mol, 273.15 K: $\Delta S = \frac{-6010\ \text{J}}{273.15\ \text{K}} = -22.0\ \text{J K}^{-1}$ | $\Delta S_{surr} = +22.0$
- **Trouton** $\Delta_{vap}S \approx 85$ (80–90) $\gg \Delta_{fus}S \approx 8$–$25\ \text{J K}^{-1}\text{mol}^{-1}$ | $S_{m,g}^\circ > S_{m,l}^\circ$
- $T_h > T_{trs}$: $\Delta S_{tot} = \Delta_{trs}H\left(\frac{1}{T_{trs}} - \frac{1}{T_h}\right) > 0$ | supercooled: $\Delta S_{tot} \approx L\left(\frac{1}{T} - \frac{1}{T_m}\right)$
- **Reaction** $\Delta_r S^\circ = \sum \nu S_m^\circ(\text{prod}) - \sum \nu S_m^\circ(\text{react})$ | $S_m^\circ(\text{element}) \neq 0$
- $Q = Pt = 100\ \text{W} \times 86\,400\ \text{s} = 8.64\ \text{MJ}$ | $\Delta S_{surr} = \frac{8.64\times10^6}{293.15} = +29.5\ \text{kJ K}^{-1}$

Third law · residual entropy 第三定律

- **Perfect crystal** $T \to 0$: $S \to 0\,(W = 1)$
- $S_m^\circ(T) = S_{res} + \sum_{\text{phases}}\int\frac{C_{p,m}}{T}dT + \sum\frac{\Delta_{trs}H}{T_{trs}}$ (area under C_p/T vs T)
- **Debye** $C = aT^3 \Rightarrow S = \int_0^T aT'^2dT' = \frac{1}{3}aT^3 = \frac{1}{3}C$ | $C = aT^r \Rightarrow S = C/r$
- $S_{res} = k_B\ln W_0$ | **ice** $W_0 \approx (\frac{3}{2})^N$: $R\ln\frac{3}{2} = 3.4$ | **CO** $W_0 = 2^N$: $R\ln 2 = 5.76$ (exp. 4–5) | 3 orient.: $R\ln 3 = 9.13\ \text{J K}^{-1}\text{mol}^{-1}$

Free energy · spontaneity · max work 自由能

- $A \equiv U - TS$ | $G \equiv H - TS = A + pV$
- $dU = TdS - pdV$ | $dH = TdS + Vdp$ | $dA = -SdT - pdV$ | $dG = -SdT + Vdp$
- **Const** T, V $\Delta A = -T\Delta S_{tot} \leq 0$ | **const** T, p $\Delta G = -T\Delta S_{tot} \leq 0$ (< 0 spont., $= 0$ eq.)
- $\Delta G = \Delta H - T\Delta S_{sys} = -T(\Delta S_{sys} + \Delta S_{surr})$ | **const** S, p $\Delta H \leq 0$
- **Max work** $W_{out,max} = -\Delta A$ (const T, total) | $W_{non\text{-}pV,max} = -\Delta G$ (const T, p) | $w_{on,rev} = \Delta A$
- $\Delta G - \Delta A = \Delta(pV) = nR\Delta T$ (ideal gas) | $\Delta T = 0 \Rightarrow \Delta G = \Delta A$
- $\Delta_r G = \Delta_r G^\circ + RT\ln Q$ | $dG = \Delta_r G\,d\xi$
- $(\Delta H, \Delta S)$: $(-,+)$ all T | $(+,-)$ none | $(-,-)$ $T < \Delta H/\Delta S$ | $(+,+)$ $T > \Delta H/\Delta S$

Engines · refrigerators · heat pumps 热机与制冷

- $\eta = \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h} \leq \eta_C = 1 - \frac{T_c}{T_h}$ (K) | $W = Q_h - Q_c$ | **rev** $\frac{Q_h}{T_h} = \frac{Q_c}{T_c}$
- $\left(\frac{\partial\eta_C}{\partial T_h}\right)_{T_c} = \frac{T_c}{T_h^2} > 0$ | $\left(\frac{\partial\eta_C}{\partial T_c}\right)_{T_h} = -\frac{1}{T_h} < 0$ | $T_c > 0 \Rightarrow \eta_C < 1$
- 473.15/293.15 K: $\eta_C = 0.380$ | (600, 300) K: 0.50 | (900, 600) K: 0.333
- **Cycle, 1 reservoir** $\Delta S_{tot} = -\frac{Q_h}{T_h} < 0$ (forbidden) | **2 reservoirs** $-\frac{Q_h}{T_h} + \frac{Q_c}{T_c} \geq 0$
- **Refrigerator** $COP_{cool} = \frac{Q_c}{W} = \frac{T_c}{T_h - T_c}$ | $W_{in} = Q_h - Q_c$ | $\Delta S_{tot} = \frac{Q_h}{T_h} - \frac{Q_c}{T_c} \geq 0$
- **Heat pump** $COP_{warm} = \frac{Q_h}{W} = \frac{T_h}{T_h - T_c} = COP_{cool} + 1$